

SHIV KUMAR

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EDUCATION

Vellore Institute of Technology

5 Year Integrated M.Tech in Computer Science and Engineering (Computational and Data Science)

2022 – 2027

Bhopal, India

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, PySpark

Machine Learning & Deep Learning: Predictive Modeling, Feature Engineering, Model Evaluation, Regression, Tree-based Models, XGBoost, Clustering, PCA, ANN, Scikit-learn, PyTorch

Generative AI & NLP: Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, OpenAI API, ChromaDB, Vector Databases, Semantic Search

Cloud & API Development: Amazon Web Services (AWS), FastAPI

Data & CS Fundamentals: Pandas, NumPy, Data Structures and Algorithms, DBMS, Operating Systems

EXPERIENCE

Software Development Intern

Oct 2025 – Dec 2025

Imperative Code

Jaipur, India

- Designed and normalized relational schemas in MySQL for enterprise-scale management systems deployed across schools and coaching centers, ensuring scalable and maintainable data architectures.
- Performed exploratory data analysis and optimized complex SQL queries by interpreting database performance metrics, improving average query execution speed by ~30%.
- Collaborated with cross-functional development teams to implement reusable and well-documented backend components, improving operational workflow efficiency by ~25%.
- Prepared structured technical documentation and presented optimization findings, methodologies, and system performance insights to development teams and project stakeholders.

PROJECTS

Autonomous Multi-Agent AI Travel Orchestrator | Python, OpenAI API, LangGraph, Agentic Workflows, AWS 2026

- Architected a LangGraph-based multi-agent travel planning system with five specialized agents and an automated budget-validation feedback loop, preventing invalid itinerary generation through iterative agent coordination.
- Built structured LLM output parsing using LangChain PydanticOutputParser with OpenAI API, implementing deterministic fallbacks for robust JSON generation.
- Built and deployed the agentic workflow as a FastAPI application on AWS EC2, returning structured JSON itineraries through stateful LangGraph orchestration.

RAG Document Q&A and Summarization System | Python, LangChain, OpenAI API, ChromaDB, FastAPI 2026

- Developed a Document Intelligence System using LangChain, OpenAI Embeddings, ChromaDB, and FastAPI for semantic PDF question answering and summarization.
- Built a Retrieval-Augmented Generation (RAG) pipeline with PDF parsing, recursive chunking, OpenAI embeddings, and semantic retrieval.
- Implemented a Map-Reduce summarization workflow with LangChain and OpenAI to generate comprehensive document summaries, exposing upload, Q&A, and summarization through FastAPI APIs.

Loan Approval & Amount Prediction System | Python, Pandas, Scikit-learn, XGBoost, FastAPI 2026

- Built a two-stage machine learning pipeline using classification for loan approval prediction and regression to estimate sanctioned loan amounts, achieving 95.3% accuracy and 0.95 weighted F1-score.
- Performed EDA, feature engineering, preprocessing comparing Logistic Regression, Random Forest, and XGBoost using F1-score, RMSE, and MAE, achieving 95% precision, 95% recall, and 0.95 F1-score.
- Deployed the end-to-end prediction pipeline with FastAPI, exposing a REST endpoint returning loan approval decisions and predicted sanction amounts.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03) | AWS

April 2026